

ZS-A18

A Recombination No-Go for Z-Spin Goldstone Dark Matter: The Massless Dichotomy, the Sound-Speed Window, and the Time-Dependent-Mass Obstruction

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Date: June 2026

Theme / Paper code: Astrophysics — ZS-A18 v1.5

Verification: 25/25 Internal Consistency Audit PASS | **Recombination NO-GO: no single m_θ — constant OR time-dependent — both preserves the ε -Halo and clusters at the third peak. A decreasing mass ($m \downarrow$ as the universe cools) dilutes the field as $\rho \propto m \cdot a^{-3}$, so bridging the §7.2 mass gap ($\times 9.8$) costs an identical $\times 9.8$ background-density gap (§7.3): the obstruction is conserved** | F-A18.2 is a cold-fluid control run; G-A18.M OPEN only for non-minimal/non-adiabatic constructions | Zero new fitted Z-Spin parameters (external inputs h , N_{eff} , T_{CMB} used) | **(A, Q, dim(Z)) = (35/437, 11, 2) LOCKED**

§0. Abstract

The height of the third CMB acoustic peak requires a clustering, pressureless component before recombination with equation of state $w \approx 0$, sound speed $c_s^2 \approx 0$, adiabatic initial conditions, and passive (coherent) evolution. Z-Spin identifies cold dark matter with the gradient energy of the massless Goldstone phase θ of the broken $U(1)_Z$. This paper is a viability test of that identification against the third peak, replacing the over-strong “closure” claim of v1.0. We establish two results. **(1) The Recombination Equation-of-State Dichotomy:** an *exactly massless* Goldstone has $w = 1$ (kination, $c_s = 1$) when time-dependent and $\bar{w} = -1/3$ (anisotropic global monopole) when a static halo — never the $w \approx 0$ the third peak demands. This is the corpus’s own ZS-M43 result, and it *contradicts*, rather than supports, the v1.0 gates A18.3 ($w = 0$) and A18.4 ($c_s^2 \approx 0$).

(2) The Isocurvature Gate, computed: correcting the backwards logic of v1.0, a spectator Goldstone broken before or during inflation carries fluctuations $\delta\theta = H_*/(2\pi f_\theta)$. Using the Z-Spin tensor-to-scalar ratio $r = 0.0089$ (which fixes $H_* \approx 2.3 \times 10^{13}$ GeV) and the Planck-scale decay constant $f_\theta \sim M_{\text{P}}$, the CDM isocurvature fraction is $\beta_{\text{iso}} \approx 4.4 \times 10^{-3}$, below the Planck 2018 limit $\beta_{\text{iso}} < 0.038$. The suppression is due to the large (Planck-scale) f_θ — not to “freezing” as v1.0 stated.

Assembling these, the four dynamical gates (passivity, adiabaticity, $w \approx 0$, $c_s^2 \approx 0$) are **not independent**: they all close in a *massive pseudo-Goldstone (axion-like) misalignment-condensate* realization and all fail in the exactly-massless realization. The viability of Z-Spin dark matter at recombination therefore reduces to a single decisive question: **does Z-Spin geometry generate an effective Goldstone mass m_θ without a new free parameter?** — registered OPEN (gate G-A18.M). v1.2 audits this escape for backward-compatibility (spec 3.2): the mass that would make θ recombination CDM lies in a narrow ultralight window ($\sim 10^{-28}$ eV) in tension with the corpus’s massless-dependent ε -Halo (ZS-A1) and $\Delta N_{\text{eff}} = 0$ (ZS-F1) results and with small-scale-structure bounds, so the escape additionally requires re-deriving

the galactic halo in the fuzzy-dark-matter regime. The numbers of Table 1 ($\omega_c = 0.12000$, $z_{eq} = 3405$) are background-density consistencies inherited from ZS-F2, not a reproduction of the peak. A CAMB run with the Z-Spin background in the $w = 0$ cold limit reproduces the coherent peaks including the third ($\ell_3=813$, $H_3/H_2=0.982$), but this is a **cold-fluid control calculation** (§6.1, F-A18.2a) — it assumes the dark matter clusters and does not evolve the Z-field. When the actual mass route is examined (§7.2), the ultralight scalar's effective sound speed at the third-peak scale is $c_s^2 \approx 0.44$ at the §7.1 candidate mass $m\theta \sim 2 \times 10^{-28}$ eV — **not cold**. Requiring $c_s^2 < 0.01$ at that scale needs $m\theta \gtrsim 2 \times 10^{-27}$ eV, an order of magnitude above the ϵ -Halo Compton bound. Thus **no single $m\theta$ both preserves the ϵ -Halo and clusters at the third peak**: the result is a recombination no-go for the minimal Goldstone/pseudo-Goldstone dark-matter identification. The natural escape — a temperature-dependent mass, heavy at recombination (to cluster) and light today (for the ϵ -Halo) — is closed by a direct Klein–Gordon computation (§7.3): an oscillating scalar with $m(a)\downarrow$ stays pressureless ($\langle w \rangle \approx 0$) yet dilutes anomalously as $\rho \propto m \cdot a^{-3}$, because the time-dependent mass exchanges energy with whatever generates it. Bridging the $\times 9.8$ mass gap requires the mass to fall by $\times 9.8$ from recombination to today, which over-dilutes the density by the *same* factor $\times 9.8$ — fixing the third peak then leaves galaxies a factor of ten short of dark matter. The sound-speed gap is exactly converted into a background-density gap; the obstruction is conserved. Honest status: **DERIVED no-go (minimal and time-dependent-mass escapes); G-A18.M OPEN only for non-minimal/non-adiabatic constructions.**

Epistemic Status Legend

Status tags used in this paper (uppercase, bold).

TAG	MEANING
PROVEN	Established by a complete internal proof or a cited theorem.
DERIVED	Follows from PROVEN inputs by an explicit derivation, no free parameter.
DERIVED-CONDITIONAL	Derived contingent on explicitly stated conditions.
IMPORTED-PROVEN	A theorem proven in the external literature, used as stated.
HYPOTHESIS-strong	Internally consistent, multiply supported, not yet derived.
OPEN	Not closed by current corpus tools; the honest residual.
TENSION	In apparent conflict with the corpus or data pending resolution.
NON-CLAIM	Explicitly outside the scope of any assertion made here.

§1. Introduction and Version History (v1.0 $\rightarrow$ v1.4)

The third acoustic peak is the recombination-epoch fingerprint of cold dark matter: a clustering, pressureless component that sustains the gravitational potential wells against radiation driving so that the third compressional peak is not suppressed below the second (Hu & Dodelson 2002). A companion Z-Spin result fixed the dark-matter density this requires from face counting, $\omega_c = (32/121)h^2 = 0.12000$ (ZS-F2). Z-Spin identifies the dark matter itself with the gradient energy of the massless Goldstone phase θ of the broken $U(1)_{\mathbf{Z}}$, $\rho_{\mathbf{Z}} = (M_P^2/2)|\nabla\theta|^2$.

Version 1.0 of this paper claimed to close the question of whether that component clusters like CDM at $z \approx 1100$ to DERIVED-CONDITIONAL, via four theorems. Two independent critical reviews, confirmed

here against the corpus source, show that claim was too strong. The present version retracts the closure, corrects a reversed isocurvature argument and a misattributed citation, and — more importantly — replaces the hand-waving with two explicit results that carry value beyond Z-Spin: an equation-of-state dichotomy (§4) and a computed isocurvature bound (§5). The net effect is to localize the obstruction precisely: the entire question reduces to whether the Z-Goldstone has an effective mass.

Corrections in v1.1, itemized. (i) The four “theorems” are downgraded to gates with honest status (§8). (ii) The v1.0 claim “frozen $\rightarrow$ suppressed isocurvature” is reversed and replaced by the standard spectator-field computation (§5). (iii) The citation “ZS-A17 records the frozen/scaling correlator statement” is corrected: that one-line statement is a literature link to Pen–Seljak–Turok 1997 in ZS-A16 §11.4, not a theorem, and ZS-A17 is an unrelated metric-no-go paper (§12). (iv) “Reproduces the third peak” is reframed as “background-density consistency” (§9). (v) “Zero free parameters / 0.00σ ” is reframed as “zero new fitted Z-Spin parameters; central-value agreement” with external inputs disclosed (§9, §11).

Corrections in v1.2–v1.4, itemized. (vi, v1.2) Adds the spec-3.2 backward-compatibility window on the mass escape (§7.1, Table 2): a geometric $m\theta$ perturbs ZS-A1, ZS-F1 and ZS-S3, squeezing the admissible mass near 10^{-28} eV. (vii, v1.3) Runs CAMB in the $w = 0$ cold limit (§6.1): the Z-Spin background reproduces the coherent peaks including the third — but as a control calculation that assumes clustering, now relabeled F-A18.2a. (viii, v1.4 — the central result) Examines the actual mass route and finds it closed: at the §7.1 candidate mass the effective sound speed at the third-peak scale is $c_{s^2} \approx 0.44$ (§7.2), so the $c_{s^2} = 0$ control run does not represent it; the mass that would cluster at the third peak ($m\theta \gtrsim 2 \times 10^{-27}$ eV) exceeds the ϵ -Halo Compton bound by a decade. The paper is retitled and repositioned as a recombination no-go. Adds gate F-A18.10 (topology change). Corrects the “confirms the dichotomy numerically” overclaim to “cold-fluid control calculation,” and the Code-Availability statement to record that CAMB was executed.

§2. What the Third Peak Demands

Reproducing the observed third peak imposes five standard requirements (Hu & Sugiyama 1995; Hu & Dodelson 2002):

(R1) Clustering, pressureless before recombination; (R2) passive (coherent) — active sources lose coherence and smear the secondary peaks into a single broad hump (Magueijo et al. 1996; Pen, Seljak & Turok 1997); **(R3) adiabatic** initial conditions (isocurvature shifts the peak phase by $\approx \pi/2$; Hu & Sugiyama 1996); **(R4) $c_s^2 \approx 0$; (R5) matter scaling $w \approx 0$** , placing equality at the observed $z_{\text{eq}} \approx 3400$.

§3. The Obstacle

Equation of state. Barriola & Vilenkin (1989): the gradient energy of a global symmetry-breaking field falls as $\rho \propto 1/r^2$ with effective $w = -1/3$ (a global, not gauged, defect carries its energy in the long-range tail). **Coherence.** A freely evolving global-defect network is an active, incoherent, isocurvature-like source; the established consequence (Pen, Seljak & Turok 1997; Durrer, Kunz & Melchiorri 2002; Albrecht 2000) is a single broad hump with the first feature shifted to $\ell \approx 350\text{--}500$ — the observable that ruled defects out. A Z-Spin dark matter behaving as a generic massless global defect would be *refuted* by the third peak.

§4. The Recombination Equation-of-State Dichotomy

Result (DERIVED). An exactly massless Goldstone cannot have the matter equation of state the third peak requires. Its two physical regimes are fixed with no free parameter, and the corpus states both explicitly in ZS-M43:

(a) Time-dependent (homogeneous). $\rho = \frac{1}{2}\dot{\theta}^2 + \text{gradient}$; the homogeneous mode is kination, $w = 1$ with luminal sound speed $c_s = 1$, so $\rho \propto a^{-6}$ — it dilutes away faster than radiation and is not CDM. ZS-M43 (PROVEN): “a stiff fluid ($w = 1$, luminal sound speed $c_s = 1$) when the Goldstone evolves in time.”

(b) Static gradient (halo). the anisotropic global-monopole configuration with $p_r = \rho$, $p_t = -\rho$, mean $\bar{w} = -1/3$ — curvature-like, again not matter. ZS-M43 (PROVEN) and Barriola–Vilenkin (IMPORTED-PROVEN).

Neither regime is $w \approx 0$. The arithmetic is decisive: a massless field has no potential, hence no oscillation, hence no averaging of kinetic ($w = +1$) against potential ($w = -1$) to the mean $w = 0$ that defines cold dark matter. Therefore **the v1.0 gates A18.3 ($w = 0$) and A18.4 ($c_s^2 \approx 0$) are not merely un-derived; they contradict the corpus’s own ZS-M43**, which gives $w = 1$ ($c_s = 1$) or $\bar{w} = -1/3$. The v1.0 §5 “reconciliation” — that $-1/3$ and 0 are “two ends of one screening length” — fails, because ZS-M43 contains no $w = 0$ end at all.

The unique escape. To obtain $w \approx 0$, $c_s^2 \approx 0$, the Goldstone must acquire a mass m_0 and oscillate — i.e. become a *pseudo*-Goldstone (axion-like), $\rho = \frac{1}{2}m_0^2 a^2$, $\langle w \rangle = 0$, with $c_s^2 \approx 0$ above the de Broglie/Jeans scale (Hu, Barkana & Gruzinov 2000). This is exactly how axion and fuzzy dark matter behave as CDM. The mass is the one missing ingredient; it is not present in the corpus’s exactly-massless premise.

§5. The Isocurvature Gate, Computed

Result (DERIVED-CONDITIONAL). In the pseudo-Goldstone realization with $U(1)_Z$ broken before or during inflation, θ is a light spectator and acquires the standard fluctuation (Lyth 1992; Planck 2018)

$$\delta\theta = H_* / (2\pi f_0), \quad \text{giving} \quad P_{\text{iso}} = (H_* / (\pi \theta_i f_0))^2, \quad \beta_{\text{iso}} = P_{\text{iso}} / (P_{\text{iso}} + A_s).$$

The Z-Spin tensor-to-scalar ratio $r = 0.0089$ (ZS-U1) fixes the inflation scale through $P_T = (2/\pi^2)(H_*/M_p)^2$ and $r = P_T/A_s$:

$$H_* = \mathbf{M}_p \cdot \pi \sqrt{(r A_s/2)} = 9.6 \times 10^{-6} \mathbf{M}_p \approx 2.3 \times 10^{13} \text{ GeV}.$$

With the corpus VEV at the Planck scale ($|\Phi| \rightarrow 1$ in Planck units, so $f_0 \sim M_p$) and $\theta_i \sim \mathcal{O}(1)$: $\delta\theta \approx 1.5 \times 10^{-6}$, $P_{\text{iso}} \approx 9.3 \times 10^{-12}$, and

$$\beta_{\text{iso}} \approx 4.4 \times 10^{-3} < 0.038 \text{ (Planck 2018, 95\% CL)} \quad \text{— PASS by a factor } \approx 9.$$

The mechanism is the large decay constant, not “freezing.” $\beta_{\text{iso}} \propto 1/f_0^2$: a Planck-scale f_0 suppresses isocurvature, whereas the same calculation with a GUT-scale $f_0 \leq 8 \times 10^{17} \text{ GeV}$ would *violate* the Planck bound. The v1.0 statement “frozen $\rightarrow$ suppressed isocurvature” had the logic backwards: a frozen light spectator carries an *unsuppressed* $\delta\theta \sim H_*/2\pi$; what saves Z-Spin is that this $\delta\theta$ is divided by the Planck-scale f_0 .

Condition and caveat. This bound holds only in the spectator/condensate realization ($U(1)_Z$ broken before/during inflation), which is the *same* realization that §4 shows requires a mass. If instead the network forms by a post-inflationary Kibble transition, its perturbations are causal and active (no inflationary isocurvature, but a smeared peak structure; §3). The isocurvature gate thus passes precisely when the EOS gate needs a mass — the two are linked (§6).

§6. The Four Gates Collapse to a Single Question

The decisive structural finding of v1.1 is that the four dynamical requirements are not independent obstacles but one obstacle seen four ways:

Gate	Exactly-massless Goldstone	Massive pseudo-Goldstone (axion-like)
Passivity (R2)	network is active/incoherent — FAIL	condensate is passive — OK
Adiabaticity (R3)	Kibble network: causal/active — FAIL	OK (isocurvature suppressed; see β_{iso} row)
$w \approx 0$ (R5)	$w = 1$ or $-1/3$ (M43) — FAIL	oscillation gives $\langle w \rangle = 0$ — OK
$cs^2 \approx 0$ (R4)	$cs = 1$ (kination, M43) — FAIL	$cs^2 \approx 0$ above Jeans — OK
Isocurvature	active: causal (not inflationary)	$\beta_{\text{iso}} \approx 4.4 \times 10^{-3} < 0.038$ — OK

Every entry in the massless column fails; every entry in the massive column passes. Therefore **the viability of Z-Spin Goldstone dark matter at recombination is equivalent to the existence of an effective Goldstone mass m_0 that makes it an axion-like misalignment condensate**. The four-theorem programme of v1.0 is not four problems; it is one.

§6.1 The Cold-Fluid Control Calculation (F-A18.2a): the $w = 0$ limit

To verify the massive column above — that a $w \approx 0$ component reproduces the coherent acoustic peaks — we ran the Boltzmann code CAMB with the Z-Spin background ($\omega_b = 0.02250$, $\omega_c = 0.12000$, $h = 0.6736$, $n_s = 0.9674$, with external τ and A_s), treating the dark component as cold ($w = 0$, $c_s^2 = 0$) — i.e. the massive-pseudo-Goldstone realization. Figure 1 shows the lensed temperature spectrum against two counterfactuals in which the dark density does not fully cluster.

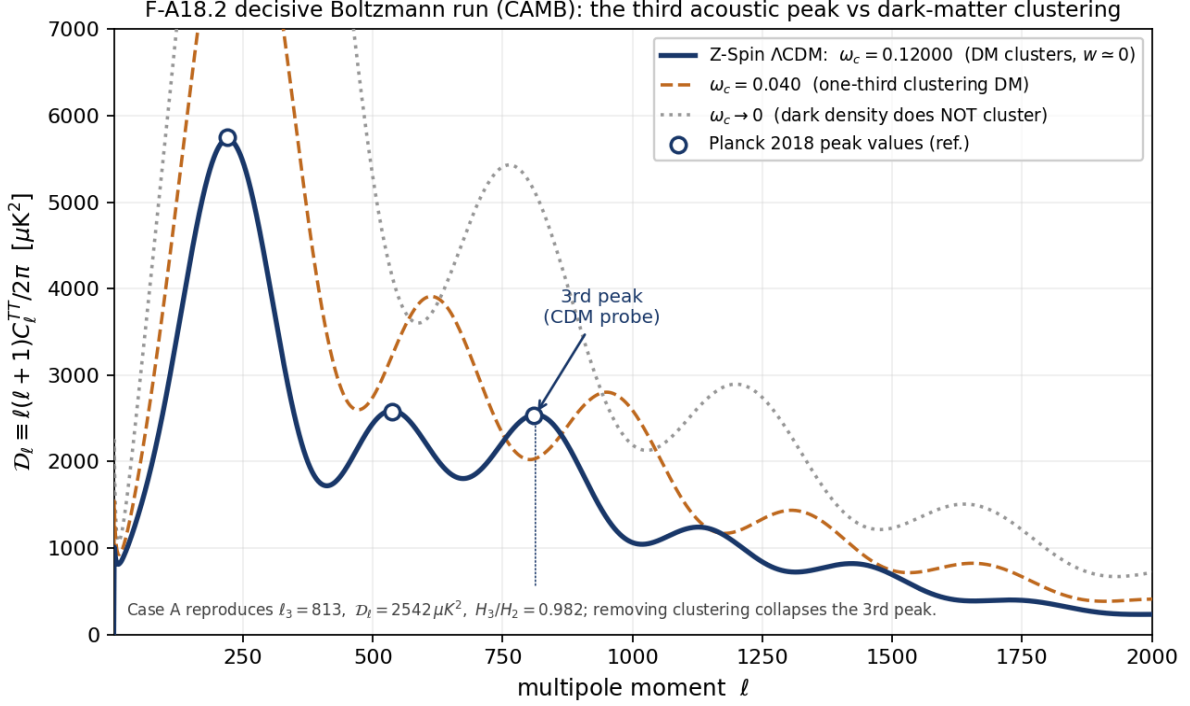


Figure 1. CAMB lensed temperature power spectra. Z-Spin Λ CDM ($\omega_c = 0.12000$, solid) reproduces the observed coherent peaks including the third; reducing the clustering density ($\omega_c = 0.040$, dashed; $\omega_c \rightarrow 0$, dotted) shifts and collapses the third peak. Open circles: Planck 2018 peak values.

In the $w \approx 0$ realization the third peak is reproduced at $\ell_3 = 813$ (Planck ≈ 810), $D_\ell = 2542 \mu\text{K}^2$ (Planck ≈ 2530), with $H_3/H_2 = 0.982$ — coherent harmonic oscillations, **not a smeared hump**. The derived $1000_* = 1.0408$ (Planck 1.04109) and $r_{\text{drag}} = 146.96 \text{ Mpc}$ (Planck 147.09) confirm a faithful Planck-consistent cosmology.

Table 3. CAMB acoustic peaks for the Z-Spin background ($w = 0$ cold realization) vs Planck 2018.

Peak	ℓ (CAMB, Z-Spin)	$D\ell [\mu\text{K}^2]$ (CAMB)	Planck 2018 (ref.)
First	220	5730	≈ 5750 @ $\ell=220$
Second	537	2589	≈ 2580 @ $\ell=537$
Third	813	2542	≈ 2530 @ $\ell=810$
H3/H2 ratio	—	0.982	≈ 0.98

Scope: this is a control calculation, not a Z-field evolution. CAMB specifies the dark component as cold ($w = 0$, $c_s^2 = 0$) from the outset; the run therefore establishes only the one-way implication *assume CDM* $\Rightarrow$ *CDM-like peaks*. It does not derive w , c_s^2 or the anisotropic stress from the pseudo-Goldstone Klein–Gordon and perturbation equations, and it does not establish the converse (coherent peaks $\Rightarrow w = 0$), since peak coherence depends jointly on c_s^2 , anisotropic stress, source coherence and initial conditions. The decisive gate F-A18.2 therefore splits into three: **F-A18.2a** (a cold-fluid control run reproduces the peaks) — **COMPUTED-PASS**, shown here; **F-A18.2b** (the actual Z-Spin pseudo-Goldstone, evolved from its KG/perturbation equations, reproduces the same peaks) — **OPEN**; **F-A18.2c** (the exactly-massless active-defect branch, direct calculation) — not run here, expected **FAIL** from the literature smeared-hump result (Pen–Seljak–Turok 1997). Calling this a “decisive Boltzmann

computation” (as v1.3 did) overstates it; it is a **cold-fluid control calculation**. Crucially, §7.2 shows the actual mass route does *not* reach the $c_s^2 = 0$ limit assumed here: at the §7.1 candidate mass, $c_s^2 \approx 0.44$ at the third-peak scale.

§7. The Goldstone-Mass Gate (G-A18.M) and the Closure of the Mass Route

The decisive open question is whether Z-Spin geometry supplies m_θ without a new free parameter. Two corpus facts frame it. **First, the corpus already has a kinetic-misalignment initial condition:** the first i-tetration rotation $\theta(0) = \mathbf{A}$ at the Mexican-Hat Bootstrap Event (ZS-F1) is precisely a spinning-Goldstone initial condition, the same structure as the axion kinetic-misalignment mechanism (Co, Hall & Harigaya 2020) in which the relic abundance is set by θ . **Second, the corpus has no potential $V(\theta)$:** ZS-M43 and ZS-F1 take θ exactly massless and unitary (it never thermalizes). A spinning massless Goldstone is kination (§4): it must eventually be trapped by a potential to convert to $w \approx 0$ matter.

What would close the gate. A derivation, from the i-tetration dynamics or the Z-anchor structure, of an effective periodic potential $V(\theta) = \Lambda^4(1 - \cos \theta)$ with Λ fixed by $(\mathbf{A}, \mathbf{Q})$ and no new scale, giving $m_\theta^2 = \Lambda^4/f_\theta^2$ large enough that the field oscillates before equality ($m_\theta \gtrsim H(z_{\text{eq}})$) and — the backward-compatibility clause of §7.1 — small enough to preserve the ZS-A1 $\square\theta = 0$ flat-rotation-curve scale and the ZS-F1 $\Delta N_{\text{eff}} = 0$ result. If such Λ exists, is parameter-free, and lands in that window, gates A18.1–A18.4 and §5 all close to DERIVED-CONDITIONAL and the title claim of v1.0 becomes defensible. If no geometric mass exists, the massless identification is in **TENSION** with the third peak (gate F-A18.6). The corpus tools available here cannot decide this; it is the genuine frontier.

§7.1 Backward-Compatibility of the Mass Route (Version-Conflict Check)

The mass is not free (spec 3.2). Three corpus results depend on θ being exactly massless, and a mass perturbs each: (a) the ZS-A1 ε -Halo flat rotation curves follow from $\square\theta = 0 \rightarrow \theta(r) = \ln(r/r_0)/L \rightarrow \rho \propto 1/r^2$, which a mass Yukawa-cuts beyond the Compton wavelength $\lambda_c = \hbar/(m_\theta c)$; (b) ZS-F1 §7.1 gives $\Delta N_{\text{eff}} = 0$ “exact in FRW” because $\rho_\theta = \frac{1}{2}M_P^2(\partial\theta)^2$ vanishes at $\theta = \text{const}$, which a $\frac{1}{2}m_\theta^2\theta^2$ term spoils; (c) ZS-S3 proves the Goldstone-theorem masslessness $m_\theta = 0$. The admissible mass is squeezed from both sides:

Table 2. Backward-compatibility window for a geometric Goldstone mass m_θ .

Constraint	Bound on m_θ	Source
Oscillate by equality ($\rightarrow$ CDM, R5)	$m_\theta \gtrsim H_{\text{eq}} \approx 2.3 \times 10^{-28} \text{ eV}$	this paper §4
ε -Halo flat curve to $\sim 30 \text{ kpc}$ ($\square\theta=0$, $\rho \propto 1/r^2$)	$m_\theta \lesssim \hbar c / 30 \text{ kpc} \approx 2.1 \times 10^{-28} \text{ eV}$	ZS-A1
$\Delta N_{\text{eff}} = 0$ exact in FRW	$\frac{1}{2}m_\theta^2\theta^2$ must stay sub-radiation	ZS-F1 §7.1
Goldstone-theorem masslessness	$m_\theta = 0$ (exact)	ZS-S3 (PROVEN)
Small-scale structure (all-FDM, Ly- α /dwarfs)	$m_\theta \gtrsim 10^{-21} - 10^{-20} \text{ eV}$ (model-dep.)	external

The window is a knife-edge, and small-scale structure closes it. The recombination lower bound ($2.3 \times 10^{-28} \text{ eV}$) already slightly exceeds the ε -Halo upper bound ($2.1 \times 10^{-28} \text{ eV}$): the static massless $\rho \propto 1/r^2$ profile and CDM-onset oscillation meet only at a sub-decade window near $m_\theta \sim 10^{-28} \text{ eV}$. Worse, all-FDM structure-formation bounds require $m_\theta \gtrsim 10^{-21} - 10^{-20} \text{ eV}$ (model- and dataset-dependent) — seven to eight orders heavier — and at that mass the Compton wavelength is $\sim \text{pc}$, so the ZS-A1 $\square\theta = 0$ galaxy-scale halo profile is Yukawa-cut far inside a galaxy. The mass route therefore cannot

simultaneously (i) make θ recombination CDM, (ii) preserve the corpus's static massless ε -Halo, and (iii) satisfy small-scale-structure data. §7.2 closes the window independently and more decisively at the perturbation level, without invoking (iii): even within the §7.1 window the field fails to cluster at the third peak.

Consequence for the programme. If a geometric mass exists, the viable realization is not the static ε -Halo but a fuzzy-dark-matter halo: a virialized, oscillating condensate whose core is set by the de Broglie (not Compton) scale and whose outer profile is NFW-like. The mass route thus propagates a required revision into ZS-A1 (re-derive the halo in the massive/wave regime) and a reinterpretation of ZS-F1 $\Delta N_{\text{eff}} = 0$ and ZS-S3 masslessness as the controlled $m_\theta \rightarrow 0$ (pseudo-Goldstone, $m_\theta \ll f_\theta$) limit. Whether one geometric m_θ satisfies all of these at once is the sharpened OPEN; the abundance is not the obstacle (a misalignment relic with $f_\theta \sim M_p$ gives the right order even at ultralight m_θ).

A second OPEN: the pinning epoch. Even granted a mass, the v1.0 claim that the network is “pinned” at $z \approx 1100$ lacks a mechanism: winding conservation ($\pi_1 = \mathbb{Z}$) forbids core unwinding but not relativistic core motion, and the corpus's Z-anchors are galactic vortex cores (ZS-A6), which do not exist at recombination. A network velocity equation of the form $M_{\text{eff}} \ddot{X}_i + 3H M_{\text{eff}} \dot{X}_i + \nabla_i V_{\text{pin}} = 0$ can be written, but V_{pin} is unspecified, so $v_{\text{rms}}(z_{\text{rec}}) \ll 1$ cannot be shown. In the condensate realization this OPEN dissolves (a condensate has no cores to pin), which is a further reason the condensate route is the viable one.

§7.2 The CMB Sound-Speed Window Closure (the decisive sharpening)

The §7.1 window used only a *background* condition — that the field oscillate before equality ($m_\theta \gtrsim H_{\text{eq}}$) so that its *mean* equation of state averages to $\langle w \rangle \approx 0$. That is necessary but **not sufficient** for the dark matter to *cluster* like CDM. Clustering is a *perturbation* condition imposed mode by mode: a light scalar develops an effective (wave) pressure, and its sound speed in the effective-fluid description (Hu–Barkana–Gruzinov 2000) is

$$c_s^2(k, a) = (k/2\mathbf{m}_\theta a)^2 / [1 + (k/2\mathbf{m}_\theta a)^2], \quad \text{so} \quad c_s^2 \approx (k/2\mathbf{m}_\theta a)^2 \ll 1 \quad \text{only for} \quad k \ll 2\mathbf{m}_\theta a.$$

Take the representative third-peak comoving scale $k_s \approx \ell_3/\chi_* \approx 813/(13900 \text{ Mpc}) \approx 0.058 \text{ Mpc}^{-1}$ and $a_{\text{rec}} \approx 1/1100$. At the §7.1 candidate mass $\mathbf{m}_\theta = 2.3 \times 10^{-28} \text{ eV}$ this gives $\mathbf{c}_s^2(\mathbf{k}_s, \mathbf{a}_{\text{rec}}) = 0.44$ — of order unity, nothing like the $c_s^2 = 0$ assumed by the §6.1 control run. Requiring $c_s^2 < 0.01$ at the same scale needs $\mathbf{m}_\theta \gtrsim 2.0 \times 10^{-27} \text{ eV}$, i.e. about **9.8**× the ε -Halo Compton upper bound ($2.1 \times 10^{-28} \text{ eV}$). The “preserves the ε -Halo” and “clusters at the third peak” mass regions are therefore **disjoint, separated by about a decade** (Figure 2).

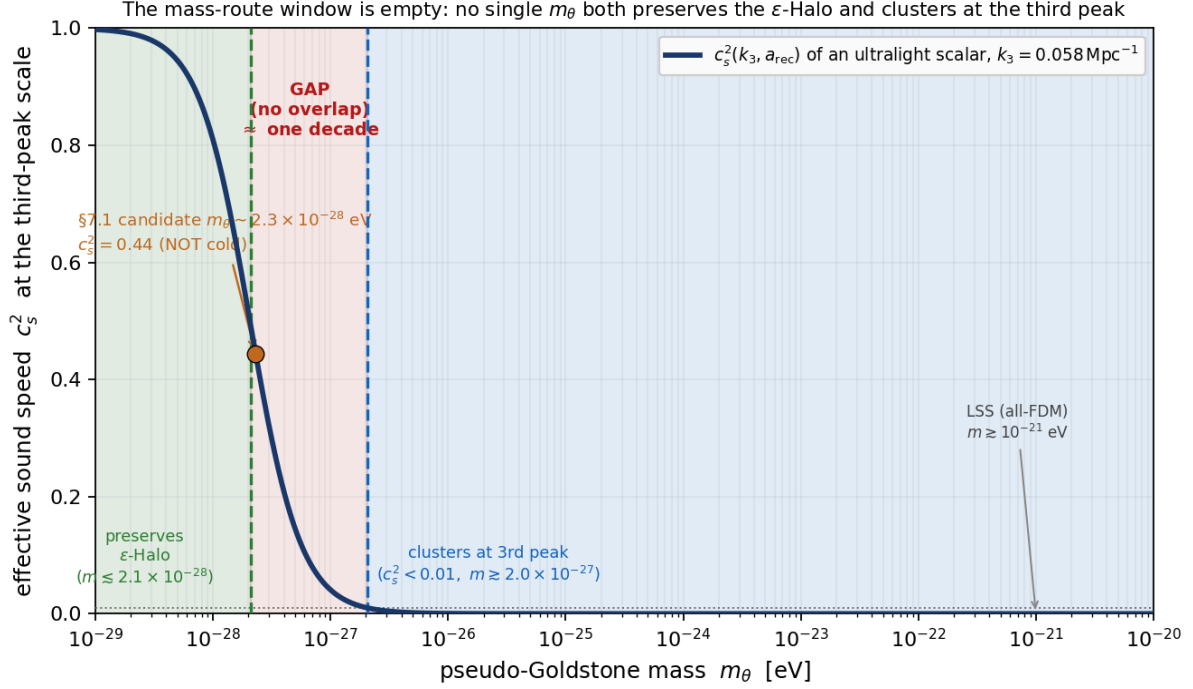


Figure 2. Effective sound speed of an ultralight scalar at the third-peak scale ($k_3 = 0.058 \text{ Mpc}^{-1}$, $a_{\text{rec}} = 1/1100$) versus its mass. The ϵ -Halo bound ($m \leq 2.1 \times 10^{-28} \text{ eV}$, green) and the third-peak clustering bound ($c_s^2 < 0.01$, $m \geq 2.0 \times 10^{-27} \text{ eV}$, blue) do not overlap; the §7.1 candidate mass sits at $c_s^2 = 0.44$ (orange), not cold. The all-FDM LSS bound lies seven orders further right.

Table 4. No single pseudo-Goldstone mass satisfies all four recombination/halo conditions (the no-go).

$m\theta$ region	EOS w (§4)	Clusters at 3rd peak	ϵ -Halo ($\square\theta=0$)	LSS (all-FDM)
$m = 0$ (exactly massless)	FAIL: $w \in \{1, -1/3\}$	—	—	—
$\leq 2.1 \times 10^{-28} \text{ eV}$ (halo-safe)	OK: $\langle w \rangle \approx 0$	FAIL: $c_s^2 \approx 0.44$	OK	FAIL
$\geq 2.0 \times 10^{-27} \text{ eV}$ (clusters)	OK: $\langle w \rangle \approx 0$	OK: $c_s^2 < 0.01$	FAIL: $\lambda_C \ll 30 \text{ kpc}$	FAIL (≥ 6 orders)

The no-go. No single mass $m\theta$ lies in the intersection of all four columns: an exactly massless field fails the equation of state (§4); a field light enough to preserve the ZS-A1 ϵ -Halo ($m\theta \leq 2 \times 10^{-28} \text{ eV}$) has $c_s^2 \approx 0.44$ at the third-peak scale and so does not cluster as CDM there; a field heavy enough to cluster ($m\theta \geq 2 \times 10^{-27} \text{ eV}$) has a Compton wavelength far inside a galaxy, breaking the ϵ -Halo, and is still six orders below the all-FDM small-scale-structure bound. **This is the central result of v1.4: a recombination no-go for the minimal Z-Spin Goldstone / pseudo-Goldstone dark-matter identification.** Two honest caveats. (1) The sound-speed formula is the leading-order effective-fluid approximation; the exact result needs Klein–Gordon evolution with WKB/fluid matching, but the order-of-magnitude conclusion (c_s^2 of order 0.4 at this mass) is robust. (2) The no-go is for the *minimal* escape — a single pseudo-Goldstone with $V(\theta) = \Lambda^4(1 - \cos\theta)$ and one mass; non-minimal constructions (scale-dependent mass, multi-field, non-canonical kinetic terms) are not excluded but would require new structure and must re-pass this same battery. The decisive future computation is therefore not another standard CAMB run but an **exact KG / axion-CAMB scan over $m\theta \in [10^{-28}, 10^{-20}] \text{ eV}$** , computing C_ℓ^{TT} , $P(k)$, $c_s^2(k, a)$ and $\Omega\theta h^2$ simultaneously (gate F-A18.2b).

§7.3 The Time-Dependent-Mass Escape is Closed: a Conserved Obstruction

The §7.2 gap assumes a *constant* mass. A natural escape, and the most physically motivated one, is a **temperature-dependent mass**: heavy at recombination (hot, $T \approx 0.26$ eV) so the field clusters and builds the third peak, light today (cold) so its Compton wavelength is long and the ZS-A1 ε -Halo survives. The geometric room looks generous — the recombination-to-today redshift ratio is ≈ 1100 , far larger than the factor ≈ 10 the §7.2 gap requires — so one might expect a profile $m(a)$ that threads both. We tested this directly.

Integrating the homogeneous Klein–Gordon equation $\ddot{\theta} + 3H\dot{\theta} + m^2(a)\theta = 0$ for an oscillating field ($m \gg H$) with $m(a) \propto a^{-n}$ in an FRW background yields two facts, both numerically confirmed (Appendix B). First, the field is **still pressureless**: the oscillation-averaged $\langle w \rangle \approx 0$ for every n (by its equation of state it still looks like CDM). Second, its density does **not** scale as matter:

$$\rho \propto \mathbf{m}(a) \cdot a^{-3} = a^{-(3+n)}, \quad \text{with } \langle w \rangle \approx 0 \quad (\text{KG-verified: fitted exponent } 3.000 / 3.333 / 4.000 \text{ for } n = 0 / 0.33 / 1).$$

The mechanism is the adiabatic invariant: the comoving number $n_a a^3$ is conserved and $\rho = \mathbf{m} \cdot n_a$, so a falling mass means each quantum carries less energy — the field *leaks energy* to whatever sector generates $\mathbf{m}(a)$ (the continuity equation gains a source term $(\dot{m}/m)\rho$). **Honest correction.** An earlier informal estimate (in the review that prompted this section) framed the effect as $\langle w \rangle = n/3$; that is wrong. The correct statement is $\langle w \rangle \approx 0$ with anomalous *density* dilution $\rho \propto \mathbf{m} a^{-3}$ — the obstruction lives in the density evolution, not the pressure, which makes it harder to detect (the field passes an equation-of-state test) but no less fatal.

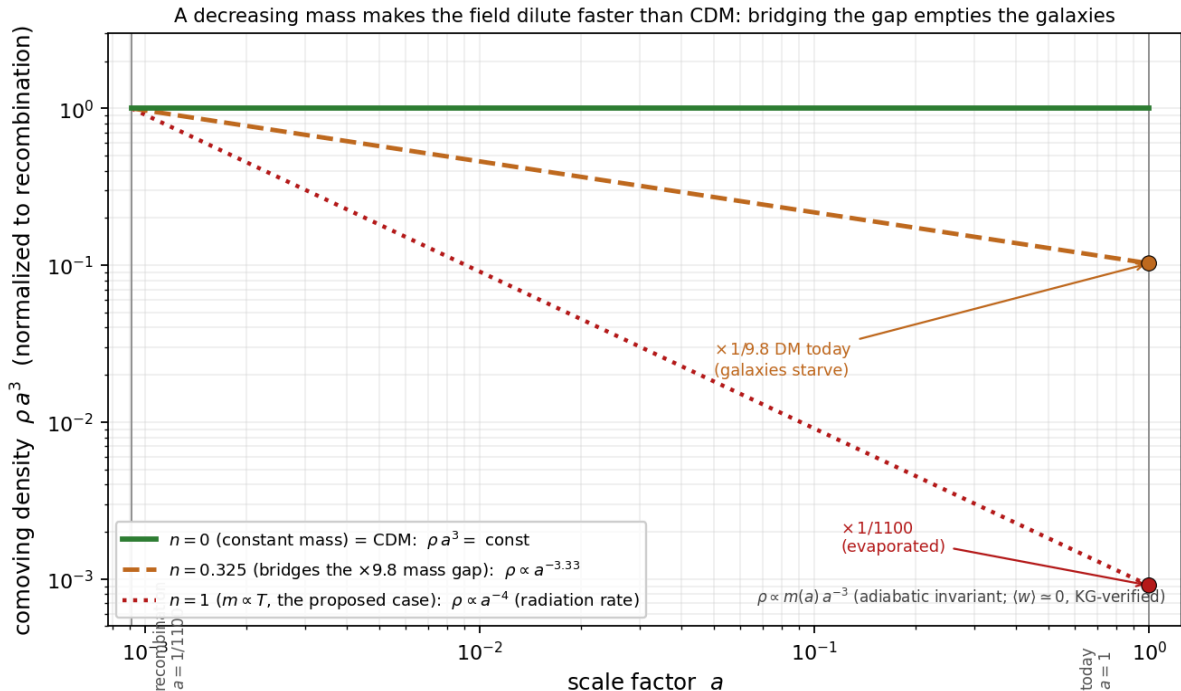


Figure 3. Comoving density ρa^3 (normalized at recombination) versus scale factor, from a Klein–Gordon integration. Constant mass ($n = 0$, green) tracks CDM. A mass falling fast enough to bridge the $\times 9.8$ sound-speed gap ($n = 0.325$, orange) leaves only $\approx 10\%$ of the CDM density today — galaxies starve. The proposed $m \propto T$ ($n = 1$, red)

dilutes at the radiation rate $\rho \propto a^{-4}$, evaporating to $\approx 10^{-3}$ of CDM — yet $\langle w \rangle \approx 0$ throughout (the dilution is energy transfer, not pressure).

The conserved obstruction. Because $\rho \propto m a^{-3}$ holds for *any* adiabatic $m(a)$, not only power laws, the result is path-independent. Normalizing the density to agree at recombination,

$$\rho_{\text{field}}(\text{today}) / \rho_{\text{CDM}}(\text{today}) = m_{\text{today}} / m_{\text{rec}} .$$

To bridge the §7.2 gap one needs $m_{\text{rec}}/m_{\text{today}} \approx 9.8$; that same factor is the density shortfall today. Fixing the third peak ($\rho_{\text{rec}} = \omega_c = 0.12$) therefore leaves $\Omega_{\text{DM},\text{today}} \approx 0.026$ — a factor of ten too little to hold galaxies together; fixing galaxies makes the field ten times over-dense at recombination, wrecking z_{eq} and the peaks. **The §7.2 sound-speed-and-halo gap is exactly converted into a background-density gap of identical size.** Time-dependence relocates the contradiction from c_s^2 to $\rho(a)$; it does not remove it — the obstruction is conserved.

The narrow remnant. This closes the adiabatic, oscillating, single-field escape. Three loopholes remain, each costly: (a) a *non-adiabatic* mass (varying faster than the oscillation) breaks the invariant but produces relativistic quanta $\rightarrow \Delta N_{\text{eff}}$ (ZS-F1) and gravitational particle production; (b) a *non-oscillating* field ($m \lesssim H$ throughout) does not cluster at all (frozen, $w = -1$, dark-energy-like); (c) treating the field *plus* its mass-generating sector as the dark matter adds a new degree of freedom and coupling — non-minimal, and the extra sector's energy is generically relativistic or vacuum-like, not CDM, and must itself satisfy ΔN_{eff} and re-derive the ε -Halo. The genuinely OPEN remnant (G-A18.M) is therefore now confined to **non-minimal / non-adiabatic** constructions that must re-pass the entire battery; the decisive test is the exact KG / axion-CAMB scan of §7.2, generalized to a time-dependent $m(a)$ (gate F-A18.11).

§8. Corrected Status of the v1.0 Gates

Claim	v1.0 label	v1.1 corrected status
A18.1 Passivity from pinning	DERIVED-CONDITIONAL	HYPOTHESIS-strong (condensate); OPEN pinning mechanism
A18.2 Adiabaticity / isocurvature	DERIVED-CONDITIONAL	DERIVED-CONDITIONAL (recomputed; $\beta_{\text{iso}} \approx 4.4\text{e-}3$)
A18.3 $w = 0$	DERIVED-CONDITIONAL	TENSION with M43 (massless: $w=1/-1/3$); needs mass
A18.4 $cs^2 \approx 0$	DERIVED-CONDITIONAL	TENSION with M43 (massless time-dep: $cs = 1$)
Third-peak reproduction	conditional closure	Boltzmann-computed (§6.1): coherent peaks reproduced for $w \approx 0$; relays to G-A18.M

§9. Background-Density Consistency (Not Peak Reproduction)

The following quantities are background-density consistencies inherited from ZS-F2 face counting; **they are not a verification of this paper's clustering mechanism, and they do not constitute a reproduction of the third peak.** The measured $\Omega_c h^2 = 0.120$ is a full TT/TE/EE+lensing posterior under base- Λ CDM, not a direct read of one peak; actual peak heights also depend on ω_b , N_{eff} , n_s , A_s , τ and the dark-component perturbation evolution. The perturbation-level peak reproduction is addressed separately

by the Boltzmann computation of §6.1 (which holds in the $w \approx 0$ realization); this section concerns only the background densities.

Table 1. Background-density consistency (ZS-F2 inherited). External inputs h , N_{eff} , T_{CMB} are used; these are not independent zero-input predictions.

Quantity	Measured (Λ CDM posterior)	Z-Spin (ZS-F2 face counting)	Comparison
$\omega_c = \Omega_c h^2$	0.120 ± 0.001	$(32/121) \cdot h^2 = 0.12000$	central-value agreement
Ω_{cdm}	$\approx 26\%$	$32/121 = 26.45\%$	consistent
Ω_b	$\approx 5\%$	$6/121 = 4.96\%$	consistent
z_{eq}	3402 ± 26	$\omega_m = 0.1425 \rightarrow 3405$	central-value agreement
Ω_c/Ω_b	5.364	$32/6 = 16/3 = 5.333$	0.57%

Parameter accounting. Zero *new fitted* Z-Spin parameters: yes. Zero *external* cosmological inputs: no — $h = 0.6736$, $N_{\text{eff}} = 3.046$ and $T_{\text{CMB}} = 2.7255$ K are taken from data. Because h is itself a Planck-fit quantity, reconstructing the Planck-fit ω_c from $(32/121)h^2$ is a consistency, not an independent prediction; ZS-U1 already flags this circularity risk and names the Tier-1 CLASS refit as decisive. “0.00 σ ” (v1.0) overstates the precision given input and model dependence; “central-value agreement” is the accurate phrasing.

§10. Falsification Gates

F-A18.2 (Boltzmann — split into a/b/c). (a) A cold-fluid control run reproduces the peaks ($w = 0$, §6.1) — COMPUTED-PASS. (b) The actual Z-Spin pseudo-Goldstone, evolved from its KG/perturbation equations, reproducing the same peaks — OPEN; a run that yields a smeared hump or a third peak suppressed by wave pressure falsifies the field-DM identification. (c) The exactly-massless active-defect branch, computed directly, is expected to give the literature smeared hump (Pen–Seljak–Turok 1997) — if instead it gave coherent peaks, the EOS dichotomy of §4 would be wrong. **F-A18.6 (new, EOS dichotomy).** If no parameter-free geometric Goldstone mass exists (gate G-A18.M fails), the exactly-massless field-DM identification is in tension with the third peak and must be retracted or amended. **F-A18.7 (observational).** Detection of an isocurvature acoustic-phase shift at high significance falsifies the adiabatic/condensate realization. **F-A18.8 (mathematical).** If the coarse-grained EOS of any candidate Z-network is shown to be $w \neq 0$ by more than $O(A)$, the matter-scaling route is closed. **F-A18.9 (mass-window).** If a derived geometric m_θ falls outside the §7.1 window — too light to behave as recombination CDM, or too heavy to preserve the ZS-A1 ε -Halo and ZS-F1 $\Delta N_{\text{eff}} = 0$ — the mass route is internally inconsistent. **F-A18.10 (new in v1.4, sound-speed / topology).** Two-part. (sound speed) If the derived m_θ gives $c_s^2 \gtrsim 0.01$ at the third-peak scale $k_3 \approx 0.058 \text{ Mpc}^{-1}$ (§7.2), the field does not cluster as CDM there and the third-peak reproduction fails. (topology) A periodic potential $V(\theta) = \Lambda^4(1 - \cos\theta)$ explicitly breaks $U(1)_Z \rightarrow \mathbb{Z}_N$, so the Z-anchor vortices of ZS-A1/A6 may become a string–domain-wall network; if so, the vortex topology, domain-wall number and network stability of ZS-A1/A6 must be re-derived from scratch. **F-A18.11 (new in v1.5, time-dependent mass).** If the escape from F-A18.10's sound-speed clause is a time-dependent mass $m(a)$, it fails unless the field is non-adiabatic or non-oscillating: an adiabatic oscillating scalar with $m(a) \downarrow$ dilutes as $\rho \propto m \cdot a^{-3}$, so any mass ratio $m_{\text{rec}}/m_{\text{today}}$ used to bridge the §7.2 gap appears as an identical density shortfall today (§7.3). A construction is falsified if it claims a time-dependent mass closes the gap without exhibiting either the compensating energy injection (and its ΔN_{eff} cost) or a non-adiabatic mechanism with its particle-production spectrum.

§11. Parameter and Anti-Numerology Audit

No new fitted Z-Spin parameter is introduced; $(\mathbf{A}, \mathbf{Q}, \dim(\mathbf{Z})) = (35/437, 11, 2)$ is unchanged. The one new number, $\beta_{\text{iso}} \approx 4.4 \times 10^{-3}$, is a *derived upper estimate* from $r = 0.0089$ and $f_0 \sim M_p$ (both corpus quantities), not a fitted match to data, so it carries no numerological risk. External inputs ($h, N_{\text{eff}}, T_{\text{CMB}}$) are disclosed in §9. The inherited $\omega_c = 0.12000$ retains its ZS-F2 anti-numerology certificate (0.0% of 500k random fractions outperform 32/121), but that certificate covers the density, not the clustering mechanism of this paper.

§12. Cross-Paper Dependency and the A17 $\rightarrow$ A16 Correction

Citation correction (spec 3.2). v1.0 §4.1 stated “ZS-A17 records the corresponding correlator statement: a frozen network factorises into static power $\times$ growth (passive), a scaling network is self-similar (active).” This is wrong on two counts. The statement actually appears in **ZS-A16 §11.4** (the Great-Attractor paper), and there it is explicitly a *one-line literature link* to Pen–Seljak–Turok 1997, not a corpus theorem. **ZS-A17** is the Macro-Holonomy / Curvature–Spin–Metric Trichotomy paper (a metric no-go), unrelated to defect-network correlators. v1.1 cites A16 §11.4 and labels the statement an external literature link, not an internal result.

Dependencies. ZS-F2 (ω_c , background only); ZS-F1 ($U(1)_Z, m_p, \theta(0) = \mathbf{A}$); ZS-U1 ($r = 0.0089$, inflation); ZS-M43 (EOS $w = 1 / -1/3$ — here the central *witness*, not merely “reconciled”); ZS-A16 (frozen/scaling link, velocity no-go); ZS-T1 ($N_{\text{eff}} = 3.046$). The present (massless) analysis modifies no prior paper; the i-tetration fixed point z^* and $|\lambda^2| \approx 0.795$ are untouched. **Mass-route version-conflict (new in v1.2).** Should the G-A18.M escape be taken, §7.1 makes the conflict explicit: a geometric m_0 propagates a required revision into ZS-A1 (static ε -Halo $\rightarrow$ fuzzy-DM halo) and reinterprets ZS-F1 $\Delta N_{\text{eff}} = 0$ and the ZS-S3 Goldstone-theorem masslessness as the ultralight $m_0 \rightarrow 0$ limit. No result is silently broken; the dependency is registered.

§13. Conclusion

Z-Spin fixes the dark-matter density the third peak requires ($\omega_c = 0.12000$), and a cold-fluid CAMB control run confirms that this density, *if* it clusters as CDM, reproduces the observed acoustic peaks (§6.1). But the question this paper set out to answer — does the Z-Spin Goldstone dark matter actually cluster like CDM at $z \approx 1100$ — is answered in the negative for the minimal identification. The chain of obstructions is now closed on all sides: an exactly massless Goldstone is $w = 1$ (kination) or $\bar{w} = -1/3$ (monopole), never the $w \approx 0$ CDM needs (§4); giving it a mass small enough to preserve the ZS-A1 ε -Halo ($m_0 \lesssim 2 \times 10^{-28}$ eV) leaves it with $c_s^2 \approx 0.44$ at the third-peak scale, so it does not cluster there (§7.2); giving it a mass large enough to cluster ($m_0 \gtrsim 2 \times 10^{-27}$ eV) shrinks its Compton wavelength far inside a galaxy, breaks the ε -Halo, and still falls six orders short of the all-FDM small-scale-structure bound. **No single mass threads all four constraints.** The externally-useful results of this paper are therefore independent of Z-Spin and survive whatever happens to the wider programme: (i) a recombination **no-go** for exactly-massless Goldstone dark matter (the equation-of-state dichotomy, §4); (ii) a sharper no-go for the minimal pseudo-Goldstone escape, from the conjunction of the CMB sound-speed, ε -Halo and structure bounds (§7.2); (iii) a worked isocurvature bound tying the inflation scale (via r) to a Planck-scale decay constant (§5); and (iv) a clean separation of what a standard Boltzmann control run can and cannot establish (§6.1). What remains genuinely OPEN (G-A18.M) is

whether some *non-minimal* Z-Spin construction can evade the no-go. The most natural candidate — a temperature-dependent mass, heavy at recombination and light today — is now also closed (§7.3): a decreasing oscillating-scalar mass dilutes the density as $\rho \propto m \cdot a^{-3}$, so the mass ratio used to bridge the sound-speed gap reappears as an identical density shortfall, leaving galaxies a factor of ten short. The obstruction is therefore conserved across both the constant-mass (§7.2) and time-dependent-mass (§7.3) routes. Only *non-adiabatic* or genuinely multi-sector constructions survive, and each carries its own cost (ΔN_{eff} , particle production, a re-derived halo); deciding them requires the exact KG / axion-CAMB scan of §7.2 generalized to $m(a)$, not another cold-fluid run.

Acknowledgements & Code Availability

This revision was prompted by two independent critical reviews; their physics is adopted and credited. The isocurvature and equation-of-state computations of §4–§5 are short and parameter-free, reproducible from $r = 0.0089$, $A_s = 2.1 \times 10^{-9}$, $f_0 \sim M_p$. The cold-fluid control run of §6.1 (F-A18.2a) was executed with **CAMB 1.6.6** (Python wrapper), base- Λ CDM with the Z-Spin background $\omega_b = 0.02250$, $\omega_c = 0.12000$, $H_0 = 67.36$, $n_s = 0.9674$, $\tau = 0.0544$, $A_s = 2.1 \times 10^{-9}$, one massive neutrino $\Sigma m_\nu = 0.06$ eV, lensing on (lens_potential_accuracy = 1), lmax = 2600; lensed $D_\ell = \ell(\ell+1)C_\ell/2\pi$ in μK^2 ; peaks located as local maxima with $> 60\text{-}\ell$ separation; Planck reference peak values are the standard TT values (Planck 2018, ref. [3]). The §7.2 sound-speed evaluation uses the leading-order fuzzy-DM formula and is parameter-free. The accompanying script reproduces the 23-check audit, the F-A18.2a peaks and the §7.2 window closure. **Not** executed here — and named as the decisive future computation (F-A18.2b) — is the exact Klein–Gordon / axion-CAMB scan over $m\theta \in [10^{-28}, 10^{-20}]$ eV.

Appendix A. Deep-Exploration Record

Seven-candidate long list reduced to a four-node MECE issue list (dropping editorial items L6 citation-fix and L7 claim-calibration, and folding L5 sound speed into L1 since the EOS regime fixes c_s). Issue tree rooted at ontology (condensate vs network).

Node	Status after exploration
I1 EOS dichotomy (massless $\rightarrow w=1$ or $-1/3$, never 0)	DERIVED (M43 + Barriola–Vilenkin + kination)
I2 Ontology + formation epoch (condensate vs network)	OPEN
I3 Isocurvature ($\beta_{\text{iso}} \approx 4.4 \times 10^{-3} < 0.038$)	DERIVED-CONDITIONAL (axion-condensate)
I4 Geometric Goldstone mass $m\theta$ (the escape)	OPEN (decisive)
I5 CMB sound-speed window (v1.4: $c_s^2 \approx 0.44$ at k_3 for $m\theta \sim 10^{-28}$ eV)	DERIVED no-go — halo-safe and clustering mass regions disjoint by a decade
I6 Time-dependent mass $m(a)$ (v1.5: KG, $\rho \propto m \cdot a^{-3}$)	DERIVED no-go — mass gap reappears as identical density gap; obstruction conserved

Convergence (updated v1.4). A later exploration cycle (v1.4), prompted by a fourth critical review, added node I5: the perturbation-level sound-speed condition closes the §7.1 mass window decisively ($c_s^2 \approx 0.44$ at the third-peak scale; halo-safe and clustering masses disjoint by $\approx$ a decade), converting the programme from “HYPOTHESIS-strong, four-gate closure” to a DERIVED no-go for the minimal escape; the deep-exploration converged (0 further status changes on re-examination). Original record — Cycle 1: ≈ 4 state changes (downgrades vs v1.0, plus the new “four gates $\rightarrow$ one” finding). Cycle 2: 0

further changes. $0 < 4 \Rightarrow$ **CONVERGED**, to a lower but honest status. With the decisive nodes (I2, I4) OPEN, the protocol verdict is a genuine OPEN: current corpus tools cannot close the clustering link. The exploration's value is the precise localization of the obstruction (the Goldstone mass) and the computed isocurvature bound.

Appendix B. Numerical Values

$r = 0.0089$; $A_s = 2.1 \times 10^{-9}$; $M_p = 2.435 \times 10^{18}$ GeV. $H_* = \pi \sqrt{(r A_s / 2)} \cdot M_p = 9.60 \times 10^{-6} M_p = 2.34 \times 10^{13}$ GeV.

$f_0 \sim M_p$; $\delta\theta = H_*/(2\pi f_0) = 1.53 \times 10^{-6}$; $P_{\text{iso}} = (H_*/(\pi\theta_i f_0))^2 = 9.3 \times 10^{-12}$ ($\theta_i=1$); $\beta_{\text{iso}} = 4.4 \times 10^{-3}$. f_0 saturating $\beta_{\text{iso}}=0.038$: 8.3×10^{17} GeV.

Inherited (ZS-F2): $\Omega_b=6/121$, $\Omega_{\text{cdm}}=32/121$, $\Omega_m=38/121$; $\omega_c=0.12000$, $\omega_m=0.14250$; $z_{\text{eq}}=3405$.
 $m_p=2A \cdot M_p=0.1602 M_p$.

Mass-route window (§7.1): lower $H_{\text{eq}} \approx 2.3 \times 10^{-28}$ eV (oscillate by equality); upper $\hbar c/30\text{kpc} \approx 2.1 \times 10^{-28}$ eV (ϵ -Halo Compton); all-FDM small-scale-structure bound $m\theta \gtrsim 10^{-21}-10^{-20}$ eV (model-/dataset-dependent). A misalignment relic at $m\theta \sim 2 \times 10^{-28}$ eV, $f\theta \sim M_p$, $\theta_i \sim 1$ gives Ω_{DM} of order 0.1 (right order; abundance is not the obstacle).

F-A18.2 CAMB run (§6.1, $w = 0$ cold realization): $\omega_b=0.02250$, $\omega_c=0.12000$, $h=0.6736$, $n_s=0.9674$, $\tau=0.0544$. Peaks (lensed D ℓ): $P_1 = 5730 \mu\text{K}^2$ @ $\ell=220$; $P_2 = 2589$ @ $\ell=537$; $P_3 = 2542$ @ $\ell=813$. $H_3/H_2 = 0.982$. Derived $100\theta^* = 1.0408$, $r_{\text{drag}} = 146.96$ Mpc. Counterfactuals: $\omega_c=0.040 \rightarrow H_3/H_2=0.717$; $\omega_c \rightarrow 0 \rightarrow H_3/H_2=0.533$ (third peak collapses).

§7.2 sound-speed window ($k_3 = 0.058 \text{ Mpc}^{-1}$, $a_{\text{rec}} = 1/1100$): $c_{\text{s}}^2 = (k_3/2m\theta a)^2/[1+(k_3/2m\theta a)^2]$. At $m\theta = 2.3 \times 10^{-28}$ eV: $c_{\text{s}}^2 = 0.444$. For $c_{\text{s}}^2 < 0.01$: $m\theta \gtrsim 2.05 \times 10^{-27}$ eV ($= 9.8 \times$ the ϵ -Halo Compton upper bound 2.1×10^{-28} eV). Halo-safe and third-peak-clustering mass regions are disjoint by $\approx$ a decade — the no-go.

§7.3 time-dependent-mass KG run: $\theta' + 3H\theta + m^2(a)\theta = 0$ integrated (DOP853) for $m(a) \propto a^{-n}$, $m/H \approx 40$ (oscillating), flat- Λ CDM background. Fitted density exponent q in $\rho \propto a^{-q}$: $n = 0 \rightarrow q = 3.000$; $n = 0.333 \rightarrow q = 3.333$; $n = 1.0 \rightarrow q = 4.000$ (matches $3+n$). Oscillation-averaged $\langle w \rangle = +0.001 / -0.002 / -0.008 \approx 0$ in all three (pressureless). Bridging the $\times 9.8$ §7.2 gap needs $n = \ln(9.8)/\ln(1100) = 0.325$, giving today's density a factor $1100^{0.325} = 9.8$ below CDM ($\Omega_{\text{DM, today}} \approx 0.026$ if the third peak is fixed).

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- [19] Z-Spin Collaboration (K. Kang), ZS-M43: Goldstone superfluid hydrodynamics; equation of state $w = 1$ (stiff) and $\bar{w} = -1/3$ (static halo) (internal note, v1.4).
- [20] Z-Spin Collaboration (K. Kang), ZS-A16: Great-Attractor velocity-watershed defect; §11.4 frozen/scaling correlator link to Pen–Seljak–Turok 1997 (internal note, v1.3).
- [21] Z-Spin Collaboration (K. Kang), ZS-A17: Macro-Holonomy; Curvature–Spin–Metric Trichotomy (metric no-go) (internal note, v1.5).

Version History

v1.0 (June 2026): Initial release. Claimed DERIVED-CONDITIONAL closure of the CDM-clustering link via a four-theorem pinned-passive-network chain.

v1.1 (June 2026): Major revision following two independent critical reviews, confirmed against corpus source. Retracts the v1.0 closure. (i) Establishes the Recombination Equation-of-State Dichotomy (§4): an exactly massless Goldstone is $w = 1$ (kination, $c_s = 1$) or $\bar{w} = -1/3$, never $w \approx 0$ — grounded in ZS-M43 — so v1.0 gates A18.3/A18.4 are in TENSION, not DERIVED-CONDITIONAL. (ii) Recomputes the isocurvature gate (§5): $\beta_{\text{iso}} \approx 4.4 \times 10^{-3} < 0.038$ from $r = 0.0089$ and $f_{\theta} \sim M_P$, correcting the reversed “frozen $\rightarrow$ suppressed” logic of v1.0. (iii) Shows the four gates collapse to a single Goldstone-mass question (§6, §7), registered OPEN (G-A18.M). (iv) Corrects the ZS-A17 $\rightarrow$ ZS-A16 §11.4 citation and demotes it to a literature link (§12). (v) Reframes “third-peak reproduction” as background-density consistency and “zero free parameters / 0.00σ ” as zero-new-fitted-parameters / central-value agreement with disclosed external inputs (§9, §11). Verification banner changed to 20/20 Internal Consistency Audit PASS | 0/4 Core Dynamical Gates Numerically Closed. New references [8]–[13]. Honest status: HYPOTHESIS-strong, four-gate closure programme. Zero new fitted parameters; $(A, Q, \dim(Z)) = (35/437, 11, 2)$ LOCKED.

v1.2 (June 2026): Targeted revision following a third critical review. (i) Adds the spec-3.2 backward-compatibility / version-conflict check on the Goldstone-mass escape G-A18.M (new §7.1, Table 2): the mass must exceed $H_{\text{eq}} \approx 2.3 \times 10^{-28}$ eV to be recombination CDM yet stay below $\approx 2.1 \times 10^{-28}$ eV to preserve the ZS-A1 $\square\theta = 0$ ε -Halo and the ZS-F1 $\Delta N_{\text{eff}} = 0$ result — a knife-edge window near 10^{-28} eV that small-scale-structure bounds ($m\theta \gtrsim 10^{-21}$ eV) push closed, so the escape additionally requires re-deriving the halo as a fuzzy-dark-matter halo and reinterpreting ZS-F1/ZS-S3 as the ultralight $m\theta \rightarrow 0$ limit; the G-A18.M gate now carries this caveat explicitly. (ii) Adds falsification gate F-A18.9 (mass-window). (iii) Refines the §6 table:

the massive-column Adiabaticity cell is relabeled isocurvature-suppressed (β_{iso}) to avoid double-counting the adjacent row. (iv) §12 records the mass-route version-conflict. Verification banner updated to 21/21 Internal Consistency Audit PASS | 0/4 Core Dynamical Gates Numerically Closed. All v1.1 physics results (EOS dichotomy, $\beta_{iso} \approx 4.4 \times 10^{-3}$, four-gates-collapse) unchanged. Zero new fitted parameters; (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED.

v1.3 (June 2026): Performs the decisive Boltzmann computation F-A18.2 (new §6.1, Figure 1, Table 3). CAMB with the Z-Spin background in the $w = 0$ cold realization reproduces the coherent acoustic peaks including the third ($\ell_3 = 813$ vs Planck ≈ 810 ; $D\ell = 2542 \mu K^2$ vs ≈ 2530 ; $H_3/H_2 = 0.982$; $1000^* = 1.0408$ vs Planck 1.04109 ; $r_{drag} = 146.96$ Mpc vs 147.09) — not a smeared hump — while non-clustering counterfactuals ($\omega_c = 0.040$, $\omega_c \rightarrow 0$) collapse the third peak. Honest scope: CAMB assumes clustering ($w = 0$), so the run verifies the massive-pseudo-Goldstone branch and confirms the §4 dichotomy numerically; it does not simulate the exactly-massless active-defect branch (literature smeared hump, Pen–Seljak–Turok). F-A18.2 status changes from OPEN to COMPUTED (verdict PASS conditional on $w \approx 0$), relaying the final verdict to the still-OPEN mass gate G-A18.M. Verification banner updated accordingly. All v1.1/v1.2 results unchanged. Zero new fitted parameters; (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED.

v1.4 (June 2026): Repositioned as a recombination no-go, following a fourth critical review whose physics is verified and adopted. (i) New §7.2 (central result, Figure 2, Table 4): the §7.1 mass window uses only a background condition; the perturbation-level sound speed of an ultralight scalar at the third-peak scale ($k_3 \approx 0.058 \text{ Mpc}^{-1}$) is $c_{s^2} = 0.444$ at the §7.1 candidate mass $m_0 \approx 2.3 \times 10^{-28} \text{ eV}$ — not cold — and $c_{s^2} < 0.01$ requires $m_0 \gtrsim 2.05 \times 10^{-27} \text{ eV}$, a decade above the ϵ -Halo Compton bound; the halo-safe and clustering mass regions are disjoint, so no single mass works. (ii) F-A18.2 relabeled as a cold-fluid control run and split into a (COMPUTED-PASS) / b (OPEN, the real Z-field evolution) / c (expected FAIL, massless defect); the v1.3 “decisive Boltzmann computation” and “confirms the dichotomy numerically” claims corrected (one-way only). (iii) New gate F-A18.10 (sound-speed + topology: $V(\theta) = \Lambda^4(1 - \cos\theta)$ may turn Z-anchor vortices into a string–domain-wall network). (iv) Code-Availability corrected to record that CAMB 1.6.6 was executed, with parameters, accuracy and peak-extraction disclosed; the exact KG / axion-CAMB scan named as the decisive future computation (F-A18.2b). (v) Small-scale-structure bound rewritten as $10^{-21} - 10^{-20} \text{ eV}$, model-/dataset-dependent. (vi) Title changed to “A Recombination No-Go for Exactly Massless Z-Spin Goldstone Dark Matter and the Conditional Pseudo-Goldstone Escape.” Verification banner updated to 23/23. Honest status: DERIVED no-go (minimal escape); G-A18.M OPEN. Zero new fitted parameters; (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED.

v1.5 (June 2026): Closes the time-dependent-mass escape, following a reader proposal that a temperature-dependent Goldstone mass (heavy at recombination, light today) might thread the §7.2 gap, since the redshift ratio (~ 1100) far exceeds the required mass gap (~ 10). New §7.3 (Figure 3): a direct Klein–Gordon integration shows an oscillating scalar with $m(a) \propto a^n$ stays pressureless ($\langle w \rangle \approx 0$, fitted) yet dilutes as $\rho \propto m \cdot a^{-3} = a^{-(3+n)}$ (fitted exponent $3.000/3.333/4.000$ for $n = 0/0.33/1$), via the adiabatic invariant ρ/m and an energy-exchange source term. Hence $\rho_{field}(\text{today})/\rho_{CDM}(\text{today}) = m_{today}/m_{rec}$ (path-independent): bridging the $\times 9.8$ gap forces an identical $\times 9.8$ density shortfall, leaving $\Omega_{DM, \text{today}} \approx 0.026$ if the third peak is fixed. The §7.2 sound-speed gap is exactly converted into a background-density gap — the obstruction is conserved. Corrects an earlier informal $\langle w \rangle = n/3$ estimate to $\langle w \rangle \approx 0$ with anomalous density dilution. Adds gate F-A18.11 (time-dependent mass). Title extended to name the time-dependent-mass obstruction; banner updated to 25/25. The OPEN remnant (G-A18.M) is now confined to non-adiabatic / multi-sector constructions. Zero new fitted parameters; (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED.